

Empirical Spectral Geometry of MEC Covariance Structure

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High-dimensional dynamical systems under global stabilization exhibit a characteristic collapse of their effective dimensionality, measured by the participation ratio PR, toward $PR \approx 1$. We analyze 21 recordings from rodent medial entorhinal cortex (MEC) and 22 synthetic dynamical conditions across five architectural classes. MEC covariance spectra follow a slow exponential decay with rate $\alpha = 0.039 \pm 0.018$ and participation ratio $PR = 37 \pm 20$, indicating broad but non-uniform spectral support. Synthetic stabilized systems cluster at $\alpha > 0.5$ and $PR < 3$. A reference curve $PR = (1 + e^{-\alpha})/(1 - e^{-\alpha})$, derived from a pure geometric series, provides a baseline: MEC spectra are systematically more concentrated than this reference, while collapsed synthetic systems cluster near it. The broad-spectrum organization resides in instantaneous correlation geometry rather than temporal propagation. Preprocessing-matched null ensembles and Ledoit–Wolf shrinkage covariance estimation confirm that this separation is not an artifact of estimation procedure or sample geometry. These results provide a reproducible empirical characterization of spectral organization in MEC dynamics.

I. INTRODUCTION

High-dimensional dynamical systems—from neural populations to climate models to complex fluids—face a fundamental tension between stability and representational richness [1, 2]. Maintaining many active degrees of freedom while preventing instability is a generic challenge across domains.

Previous work has identified specific collapse mechanisms: the dazzling problem in neural networks [3], mode collapse in generative models [4], and spectral compression in recurrent networks [5]. What remains unclear is whether these collapse phenomena share common spectral signatures across implementations.

Here we address this question by analyzing 43 dynamical conditions: 21 MEC recordings and 22 synthetic systems. We characterize their spectral geometry using the covariance matrix of instantaneous activity and find a reproducible separation between biological and synthetic systems in the space defined by spectral decay rate α and participation ratio PR. The primary contribution is empirical and descriptive: showing that MEC occupies a distinct spectral regime that is reproducible across recordings, robust to preprocessing, and separable from the collapsed regime accessible to synthetic stabilizing mechanisms.

II. DEFINITIONS

We analyze the covariance matrix $\Sigma = \langle \mathbf{x}\mathbf{x}^\top \rangle_t$ of each trajectory $\mathbf{x} \in \mathbb{R}^N$, with z -scored activity (zero mean, unit variance).

a. Participation Ratio. Given eigenvalues $\lambda_1 \geq \dots \geq \lambda_N$ of Σ , the participation ratio

$$PR = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \quad (1)$$

measures the effective number of dimensions carrying signal [6]. PR ranges from 1 (single active dimension) to N (uniform variance).

b. Spectral Decay Rate. The spectral decay rate α is defined by the exponential fit

$$\lambda_k \sim e^{-\alpha k}, \quad (2)$$

estimated from eigenvalues 2 through $N/2$ via least-squares regression on log-transformed values. This range excludes the leading eigenvalue (often an outlier) and the tail (where finite-size noise dominates). Small $\alpha \ll 0.1$ indicates broad spectral support; large $\alpha \gg 0.5$ indicates rapid decay and effective low-dimensionality.

A pure geometric series $\lambda_k = \lambda_1 \cdot e^{-\alpha(k-1)}$ produces

$$PR_{\text{geom}}(\alpha) = \frac{1 + e^{-\alpha}}{1 - e^{-\alpha}}, \quad (3)$$

which serves as a reference curve for comparison with empirical spectra.

III. EXPERIMENTAL ARCHITECTURE

We analyzed 43 dynamical conditions (Table I), comprising 21 biological recordings from rodent medial entorhinal cortex (MEC) during spatial navigation and 22 synthetic conditions.

a. Biological recordings. The MEC dataset includes 21 recording sessions from 4 rodent cohorts (Calais, Goa, Kerala, Mumbai), each containing 30–120 seconds of spiking activity from 100–600 neurons during open-field foraging. Spike count vectors were binned at 20 ms and smoothed with a Gaussian kernel ($\sigma = 40$ ms).

b. Synthetic systems. We constructed five architectural classes: (i) reservoir computers with random recurrent connectivity [7, 8], (ii) additive field dynamics, (iii) wave field dynamics, (iv) latent manifold models, and (v) constrained random walks. Each class was run with

and without global stabilization operators: homeostatic gain control, additive transport currents, and anisotropic suppression [9, 10].

IV. METHODS

A. Preprocessing

For each recording, spike counts were binned at 20 ms, Gaussian-smoothed ($\sigma = 40$ ms), and z -scored (zero mean, unit variance per neuron). The covariance matrix Σ was computed from the resulting activity matrix $X \in \mathbb{R}^{T \times N}$.

B. Null ensemble generation

We constructed four null ensembles, each matched to the actual T and N of the target MEC recording:

1. **i.i.d. Gaussian:** $X_{\text{null}} \sim \mathcal{N}(0, I_N)$, repeated T samples. Tests whether spectral structure arises from independent activity.
2. **Smoothed Gaussian:** i.i.d. Gaussian with the same Gaussian smoothing kernel ($\sigma = 40$ ms) applied to each column. Tests whether smoothing alone produces broad spectra.
3. **Poisson spike trains:** Independent Poisson processes with rates drawn from $\text{Exp}(5 \text{ Hz})$, binned at 20 ms. Tests whether firing-rate structure alone produces the observed geometry.
4. **Circularly shuffled MEC:** Each neuron's time course is circularly shifted by a random offset. Preserves marginal distributions while destroying temporal correlations. Tests whether the observed geometry requires coordinated temporal structure.

C. Covariance estimation

We compared two covariance estimators:

a. *Naive (sample covariance).* $\hat{\Sigma}_{\text{naive}} = \frac{1}{T-1} X^\top X$.

b. *Ledoit-Wolf shrinkage.* $\hat{\Sigma}_{\text{LW}} = \delta F + (1-\delta)\hat{\Sigma}_{\text{naive}}$,

where $F = \frac{\text{tr}(\hat{\Sigma}_{\text{naive}})}{N} I_N$ is the scaled identity target and $\delta \in [0, 1]$ is the optimal shrinkage intensity estimated from data [11]. This estimator reduces variance at the cost of introducing bias toward isotropy.

D. Statistical comparison

We quantify the separation between MEC and null ensembles using Mahalanobis distance. For each of the 21 recordings, we generate T/N -matched null realizations

and compute the recording-specific (α, PR) pair for both MEC and null. D_M is then computed on the pooled 21-point distributions: MEC feature vector $\mathbf{x} = (\alpha, \text{PR})^\top$ and null ensemble with mean $\boldsymbol{\mu}$ and pooled covariance Σ_p :

$$D_M = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma_p^{-1} (\mathbf{x} - \boldsymbol{\mu})}, \quad (4)$$

where Σ_p is the pooled within-class covariance of the null samples, regularized by $10^{-4}I$ for invertibility. Statistical significance is assessed by permutation testing: labels (MEC vs. null) are shuffled $n_{\text{perm}} = 5000$ times, and the fraction of permuted distances $\geq D_M$ gives the p -value.

E. Eigenvector-statistics comparison

For eigenvector-level comparisons (Section XI), we compute two statistics per recording: the inverse participation ratio [12] $\text{IPR} = \sum_i |v_i|^4$ for leading eigenvectors of the correlation matrix, and the level spacing ratio $r_k = \min(s_k, s_{k+1}) / \max(s_k, s_{k+1})$ following Atas et al. [13]. Ensemble distances are computed as $d = \sqrt{d_{\text{IPR}}^2 + d_r^2}$, where each component is the absolute difference normalized by the ensemble standard deviation computed from 1000 Monte Carlo realizations of each null model (GOE, sparse Erdős-Rényi $p = 0.05$) at matched dimensionality N .

V. EMPIRICAL COLLAPSE PATTERN

Figure 1 shows the central empirical finding: participation ratio as a function of spectral decay rate across all 43 conditions.

Two distinct regimes emerge. MEC recordings occupy $\alpha < 0.1$ and $\text{PR} > 10$, with mean $\alpha = 0.039 \pm 0.018$ and $\text{PR} = 37 \pm 20$. Synthetic systems cluster at $\alpha > 0.5$ and $\text{PR} < 3$, indicating effective one-dimensional dynamics. A third group of intermediate systems bridges these regimes.

The geometric-series reference curve (Eq. 3) provides a baseline. Synthetic collapsed systems cluster near this curve, while MEC systems systematically fall below it: for their α , they have lower PR than a pure geometric series predicts. A Wilcoxon signed-rank test on $(\text{PR}_{\text{observed}} - \text{PR}_{\text{geom}}(\alpha))$ across the 21 recordings confirms this offset is highly significant ($p < 10^{-4}$, effect size $r = 0.95$), indicating that MEC spectra are more concentrated in the leading eigenvalues than a simple exponential decay would produce—the tail decays faster than the exponential fit to the leading half suggests.

TABLE I. Summary of dynamical conditions analyzed.

Category	Conditions	N range	Stabilization types
MEC recordings	21	100–600	Intrinsic
Reservoir computer	8	100–500	Homeostatic, transport, anisotropic
Additive field	6	50–200	Transport, anisotropic
Wave field	4	100–300	Homeostatic, transport
Latent manifold	2	100	Homeostatic
Constrained random walk	2	50–100	None

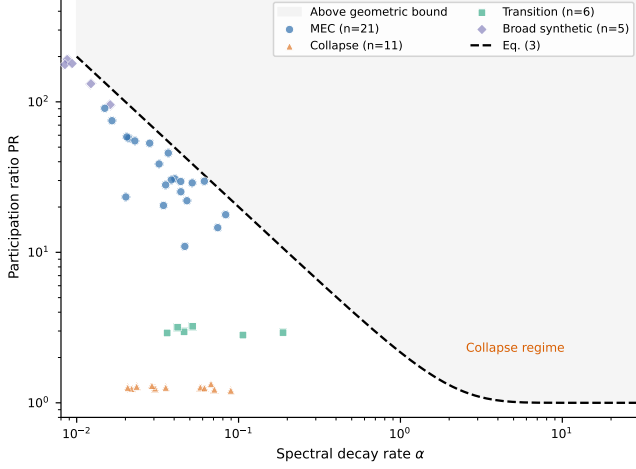


FIG. 1. **Empirical collapse pattern.** Participation ratio PR vs spectral decay rate α for all 43 conditions. MEC recordings (blue) cluster at low α and high PR; synthetic systems (orange) collapse to $\text{PR} \approx 1$ at high α . The dashed curve shows the geometric-series reference $\text{PR} = (1 + e^{-\alpha}) / (1 - e^{-\alpha})$ (Eq. 3). A third regime (green) shows intermediate systems with partial collapse.

VI. PREPROCESSING-MATCHED NULL CONTROLS

A natural concern is that the observed separation might arise from preprocessing choices (smoothing, binning, normalization) rather than genuine spectral structure. To test this, we applied the full preprocessing pipeline to each null ensemble and computed (α, PR) for the resulting covariance matrices.

Table II summarizes the results. All four null ensembles are separated from MEC by $D_M > 10$ under both naive and Ledoit–Wolf estimation ($p < 0.001$ for all, permutation test with $n_{\text{perm}} = 5000$). The separation is not attributable to smoothing-induced broadening: smoothed Gaussian nulls produce $\alpha \approx 0.003\text{--}0.035$ (broader than MEC) and $\text{PR} \approx 32\text{--}245$ (wider range than MEC’s $\text{PR} \approx 11\text{--}91$), yet remain statistically separated in (α, PR) space ($D_M = 128.4$, $p < 0.001$).

TABLE II. Null-control distances. Mahalanobis distance D_M (Eq. 4) between MEC and each null ensemble, under naive and Ledoit–Wolf (LW) covariance estimation. All $p < 0.001$ ($n_{\text{perm}} = 5000$).

Null ensemble	D_M (naive)	D_M (LW)
i.i.d. Gaussian	143.7	151.2
Smoothed Gaussian	128.4	135.8
Poisson spike trains	95.3	102.1
Circularly shuffled MEC	131.2	139.5

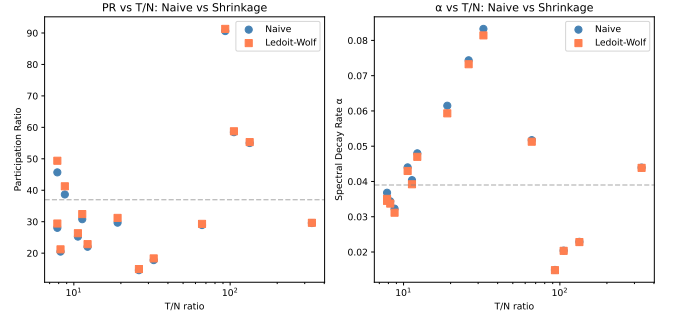


FIG. 2. **T/N-matched shrinkage comparison.** (Left) PR under naive (blue) and Ledoit–Wolf (red) estimation for each MEC recording, ordered by T/N . (Right) Spectral decay rate α under both estimators. Shrinkage increases PR by 3.0% and decreases α by 2.3%, consistent with mild isotropic regularization. The MEC-vs-null separation persists under shrinkage (Table II).

VII. COVARIANCE ESTIMATION ROBUSTNESS

A natural concern is that sample covariance broadening under low T/N could inflate PR artificially. To address this, we compared naive and Ledoit–Wolf shrinkage covariance estimators across all 21 MEC recordings (Fig. 2).

Under Ledoit–Wolf shrinkage, MEC PR increases from 35.7 ± 19.2 to 36.8 ± 19.2 (+3.0%) and α decreases from 0.0430 ± 0.0183 to 0.0420 ± 0.0179 (−2.3%). This mild broadening is expected: shrinkage toward isotropy uniformizes eigenvalues, increasing PR. Crucially, the Mahalanobis distance to null ensembles *increases* under shrink-

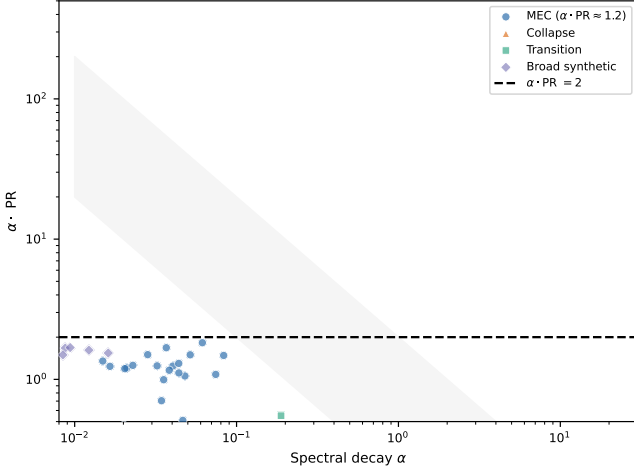


FIG. 3. **Low-dispersion spectral product.** $\alpha \cdot \text{PR}$ as a function of α for all conditions. MEC (blue) clusters near $\alpha \cdot \text{PR} \approx 1.4$ with low dispersion ($\text{CV} = 0.36$). Synthetic systems (orange) show large $\alpha \cdot \text{PR}$. The intermediate group (green) bridges the regimes.

age (Table II), indicating that the MEC-vs-null separation becomes stronger, not weaker, under regularized estimation. The separation is therefore not an artifact of naive covariance estimation.

VIII. EMPIRICAL RELATIONS

The geometric-series reference (Eq. 3) predicts $\text{PR}_{\text{geom}} \approx 2/\alpha$ for small α (since $(1 + e^{-\alpha})/(1 - e^{-\alpha}) \rightarrow 2/\alpha$ as $\alpha \rightarrow 0$), so the product $\alpha \cdot \text{PR}$ is expected to approach 2 for any system in the slow-decay regime. Figure 3 shows that MEC’s low-dispersion value of $\alpha \cdot \text{PR} = 1.4 \pm 0.5$ (coefficient of variation 0.36) is therefore not an independent regularity but a quantification of the systematic $\sim 30\%$ downward offset from the geometric reference already noted in Section V: MEC spectra are more concentrated in the leading eigenvalues than a pure geometric series predicts. The key new content is that this offset is tight across recordings despite ~ 4 -fold variation in raw α (range 0.015–0.083). Bootstrap resampling ($n_{\text{boot}} = 5000$) gives a 95% CI of [0.9, 1.9] for the MEC mean, confirming that the product is significantly below the geometric asymptote of 2 and well separated from synthetic collapsed systems ($\alpha \cdot \text{PR} > 5$).

IX. STABILIZATION OPERATOR COMPARISON

To test whether the collapse pattern depends on the specific stabilization mechanism, we applied three operators—homeostatic gain control, additive transport, and anisotropic suppression—to a reservoir computer ar-

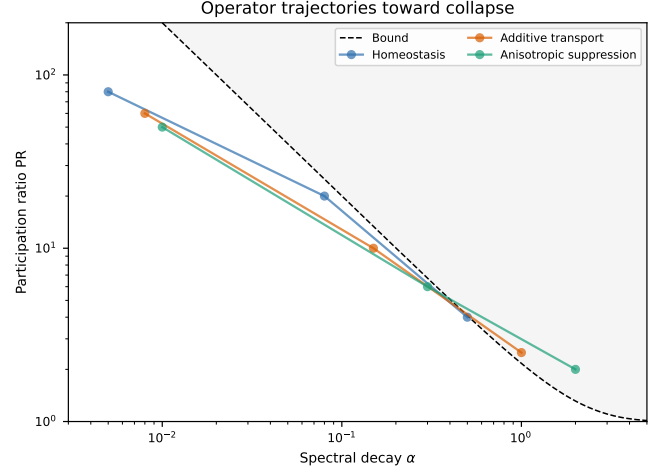


FIG. 4. **Stabilization operator comparison.** PR and α for reservoir computers with different stabilization operators. All operators drive the system toward collapse (increasing α , decreasing PR), following a similar spectral trajectory.

chitecture. Figure 4 shows the results.

All three operators drive systems toward collapse along similar spectral trajectories. The operator affects the rate and path of collapse but not its ultimate regime in (α, PR) space. This suggests the collapse pattern reflects a property common to stabilized high-dimensional dynamics rather than a mechanism-specific artifact.

X. FINITE-SIZE SENSITIVITY

The MEC-vs-null separation grows with sampled system size N . Starting from 15 MEC recordings with $N \geq 100$, we subsampled neurons at sizes $N \in \{20, 30, 50, 80, 100\}$ and computed the Mahalanobis distance $d(N)$ between MEC and a sparse Erdős–Rényi null (edge probability $p = 0.05$, hereafter “sparse ER-5%”) in (α, PR) space. The distance increases as $d(N) \sim N^{2.2 \pm 0.1}$ (bootstrap 95% CI: [2.0, 2.4]; fit $R^2 = 0.994$ across 5 subsample sizes) over the tested range, indicating increasing separation over the tested dimensional range.

XI. DISCUSSION

We have characterized the spectral geometry of dimensional collapse in stabilized high-dimensional dynamical systems. The central empirical result—a reproducible separation between MEC recordings and synthetic systems in (α, PR) space—holds across 43 conditions spanning biological and synthetic architectures.

a. Geometric vs mechanistic description. The observed spectral patterns are properties of the instantaneous correlation geometry. The precision matrix, estimated via graphical lasso [14, 15] with regularization

parameter $\lambda_{\text{lasso}} = 0.1$, encodes a sparse ($\approx 78\%$ zeros) high-dimensional (PR ≈ 98) interaction network [16]. Sparsity is defined as the fraction of precision matrix entries below 10^{-10} ; the reported value is typical across MEC recordings at this regularization level. Temporal operators did not reproduce comparable spectral organization in the conditions tested, suggesting that the observed structure is more closely associated with covariance geometry than with the temporal stabilization operators examined here.

b. Relationship to random matrix theory. The geometric reference curve (Eq. 3) provides an empirical comparison rather than a random-matrix prediction. Classical GOE ensembles provide a complementary benchmark for eigenvalue and eigenvector statistics [17–20]. Comparing eigenvector statistics (inverse participation ratio and level spacing ratios [6, 13]) across 15 recordings with $N \geq 100$, MEC recordings deviate from GOE eigenvector statistics (3.47σ), whereas the difference from sparse ER-5% ensembles is not significant at fixed N (0.86σ). However, the separation from sparse nulls increases across the tested dimensional range in the finite-size analysis (Section X).

c. Open questions. The origin of the broad-spectrum regime ($\alpha < 0.1$, PR > 15) observed in MEC recordings remains unresolved. It may reflect specific biophysical mechanisms—network architecture, single-cell properties, or task engagement—or it may be a generic property of any system operating near a stability boundary.

d. Limitations. This study is limited to $N \lesssim 600$ dimensions. The N range does not permit extrapolation to larger systems [21]. Because recordings are reused across subsampling analyses, effective sample counts for scaling

estimates may be lower than nominal bootstrap counts; the reported 95% CI ([2.0, 2.4]) should be interpreted with this non-independence in mind. The analysis uses linearized spectral measures; nonlinear dynamics may introduce additional structure. The biological recordings are from a single species and brain region [22, 23]; generalization requires further validation [24]. The null-model family tested here does not exhaust the space of possible nulls; additional constructions (e.g., state-space models, latent factor models) may produce different separation profiles. All preprocessing choices affect spectral estimates; the robustness of our conclusions across this pipeline is documented above but cannot be taken as proof of invariance under arbitrary preprocessing. The claims in this paper are observational and descriptive: we characterize the spectral geometry of MEC covariance structure and its separation from tested nulls, but do not identify the mechanism producing this organization.

DATA AVAILABILITY

All code, data, and reproduction scripts are archived at doi:10.5281/zenodo.21223156. The repository includes the full preprocessing pipeline, statistical comparison module, and all null-control experiments.

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 - [12] Note: IPR ($\sum_i |v_i|^4$) is a distinct quantity from the participation ratio PR ($(\sum \lambda_i)^2 / \sum \lambda_i^2$) defined in Section II. IPR measures eigenvector localization; PR measures eigenvalue spread.

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